

Topic 1.7 – Introductory organic chemistry

The introduction to organic chemistry shows how chemists work safely with potentially hazardous chemicals by managing risks.

1.7.1c and 1.7.1d	Students should be encouraged in their laboratory work to appreciate the difference between hazard and risk. They should appreciate that the total elimination of risk is almost impossible if society is to function normally and that laboratories are generally very safe environments compared with the risks of everyday life such as taking a car ride. Students could be encouraged to look at comparative risks associated with the use of applied pesticides compared with those from natural pesticides generated by nature. Students should understand the reasoning behind the steps that can be taken to minimise risk so that laboratory reactions can be carried out in safety. HSW 5, HSW 9
1.7.2c and 1.7.2d	Students could use the internet to research data concerning alternative fuels and compare the relative environmental impact in terms of reducing emissions and sustainability. The data could then be presented using ICT. An example might include the use of hydrogen as a fuel in motor cycles. Questions such as the carbon neutrality of hydrogen could be considered taking into account the energy used to produce the hydrogen, the problems associated with refuelling, the fact that the emissions are greenhouse gases and the myth that hydrogen is dangerous because it is 'explosive' could be exposed. HSW 8, HSW 2, HSW 3, HSW 12
1.7.2e	If linked with 1.4g several potential mechanisms could be offered and tested using bond energy and could provide evidence for the probable mechanism. This is an example of the use of a model to help chemists explain and classify reactions. HSW 1, HSW 2